

Activating the Gate-Opening of a Metal–Organic Framework and Maximizing Its Adsorption Capacity

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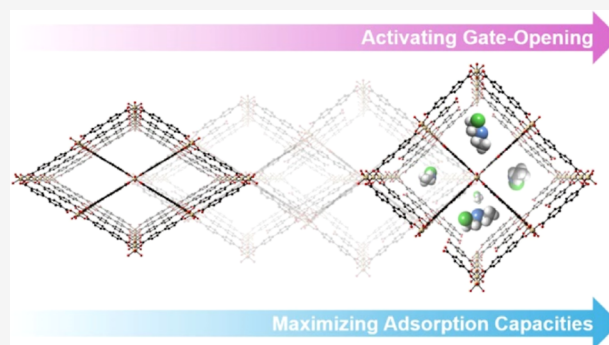


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ABSTRACT: Metal–organic frameworks (MOFs) are well-known porous materials owing to their useful adsorption properties; however, some MOFs have limited adsorption capabilities, which can significantly undermine their success as porous materials. Therefore, maximizing their porosity is critical for unlocking their full potential and expanding their practical utilization, such as gas storage, separation, and removal. In this study, flexible MOFs with defined defects were synthesized using a ligand-mixing strategy to improve their porosity and maximize their adsorption capacities. Specifically, we employed a combination of two organic linkers, 4,4'-biphenyldicarboxylic acid (H_2BPDC) and 1,4-benzenedicarboxylic acid (H_2BDC), in various ratios, to fabricate flexible In-MIL-53D hybrids containing controllable defects within the structure due to the incorporation of the short linker (H_2BDC) compared to the original linker (H_2BPDC). These structural defects in the In-MIL-53D hybrids activated their gate-openings and enhanced gas adsorption capacities for N_2 and CO_2 . Moreover, the gate-opened activated hybrids exhibited excellent adsorption capacity for the harmful chemical warfare agent simulant, 2-chloroethyl ethyl sulfide (CEES). However, excessive incorporation of defects disrupted the framework's integrity, compromising its stability and increasing the risk of collapse. Therefore, achieving an optimal level of defect incorporation is essential to balance structural stability with enhanced functionality. Among the hybrids, the sample with approximately 39% incorporation of the short linker exhibited up to an 11-fold increase in adsorption capacity for CO_2 at 1 P/P_0 . In addition, this hybrid demonstrated up to 5-fold higher CEES adsorption capacity compared to the pristine In-MIL-53D, highlighting its potential for advanced utilization in relevant fields.



INTRODUCTION

Metal–organic frameworks (MOFs) have garnered significant attention because of their exceptional porosity, high surface area, and versatile structural design, which make them invaluable for a wide range of applications, including gas storage, separation and removal, catalysis, and drug delivery.^{1–14} MOFs with outstanding adsorption properties are critically important for the selective removal of toxic chemicals that pose severe risks to human health and the environment.^{5–9} Despite their potential, many MOFs suffer from intrinsic structural limitations that restrict their adsorption capacities, thereby limiting their efficacy in applications that demand substantial porosity. Addressing these limitations and maximizing the adsorption capacity of MOFs is essential for advancing their utility in environmental and industrial applications. Recent advancements in MOF synthesis have enhanced our ability to tailor specific porosity and adsorption properties. For physisorption, an effective approach involves the incorporation of larger, bulkier ligands, which increases the pore volume and pore size.^{15–17} In contrast, chemisorption can be enhanced by introducing specific chemical sites or controlled structural defects that provide additional adsorption sites and increase pore accessibility.^{18–24} An approach known as defect engineering has been developed

that involves ligand mixing, modulation of reaction conditions, and postsynthetic modifications.^{18–24} Among these, the ligand mixing strategy has proven effective for precise control of the defect structures within MOFs, enabling fine-tuning of the overall properties of the framework.^{19–21} Flexible MOFs can adapt their conformations in response to changes in the guest molecules, pressure, or temperature and they can exist in either closed or open conformations.^{25–35} They exhibit fascinating properties. For example, flexible MOFs have shown considerable promise with adsorption capacities that adapt in response to external stimuli.^{25–35} However, a notable limitation of flexible MOFs is that they tend to remain closed after drying (Scheme 1a).^{27–29,33} This can occur due to π – π interactions between the aromatic rings of the linkers, which stabilize the closed form and require higher energy to reopen the structures.^{27–29,33}

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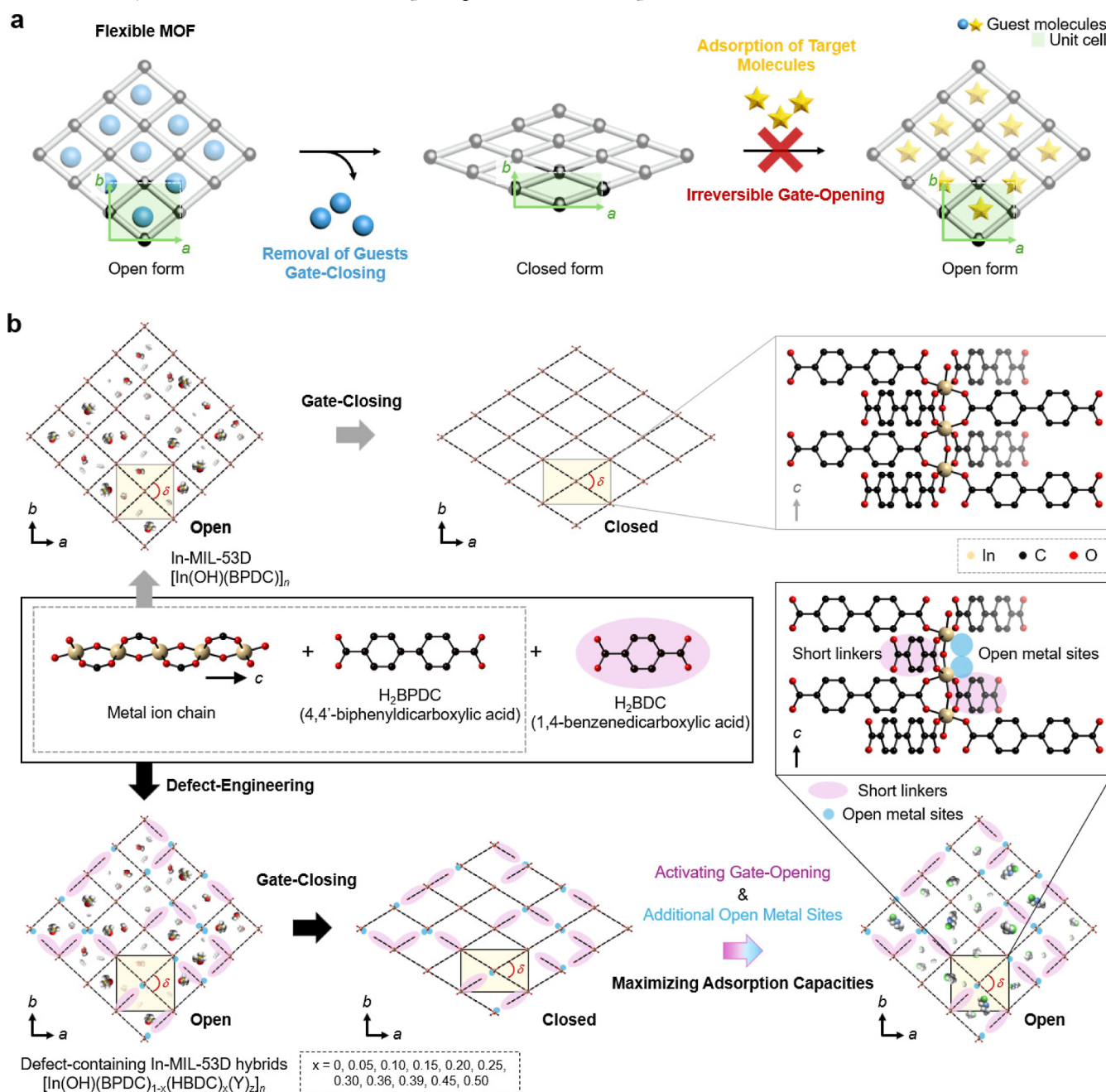
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Scheme 1. (a) Schematic of Gate-Closing and Irreversible Gate-Opening of Flexible MOF. The Metal (Black Ball), Organic Linker (Gray Rod), Guest Molecules (Blue Ball and Yellow Star), and Unit Cell Parameters (Green Box) are Illustrated, the Guest Molecules Occupy the Pores, Keeping the MOF in an Open Form. Upon Drying, the Guest Molecules are Removed, Causing the Flexible MOFs to Transition to Their Closed Form Due to Irreversible Gate-Closing, a Typical Limitation of Flexible MOFs, Exposure to Guest Molecules Does Not Reopen the Pores. (b) Schematic of the Construction of In-MIL-53D and Defect-Containing In-MIL-53D Hybrids via the Solvothermal Reaction Ball-and-Stick Representations of the 3D Orthorhombic Structure of In-MIL-53D and Hybrid In-MIL-53D with a Rhombic Lattice in the *ab* Plane. Light-Yellow Boxes Show Unit Cell Parameters. The Structures of In-MIL-53D and Hybrids Were Modified Based on Al-MIL-53D Using the Vesta Software.³⁶ In (Beige), C (Black), and O (Red); H Atoms are Omitted for Clarity. The Gray Box Shows the 1D Metal-Ion Chain in the *c* Direction of In-MIL-53D The Black Box Shows the 1D Metal-Ion Chain in the *c* Direction of In-MIL-53D Hybrids, where Pink Circles Represent the Incorporated HBDC[−] and Blue Circles Represent the Open Metal Sites. The Incorporation of HBDC[−] in In-MIL-53D Hybrids Shows Activated Gate-Opening with Additional Open Metal Sites



Ramsahye et al. addressed this limitation by synthesizing MIL-88, a well-known flexible MOF. They incorporated organic spacers within pristine organic linkers to reduce $\pi-\pi$ interactions between the phenyl groups, enabling the MOF to

stay in the open form.²⁹ Overcoming these typical limitations of flexible MOFs is essential before their potential for applications such as adsorption and removal can be fully harnessed.

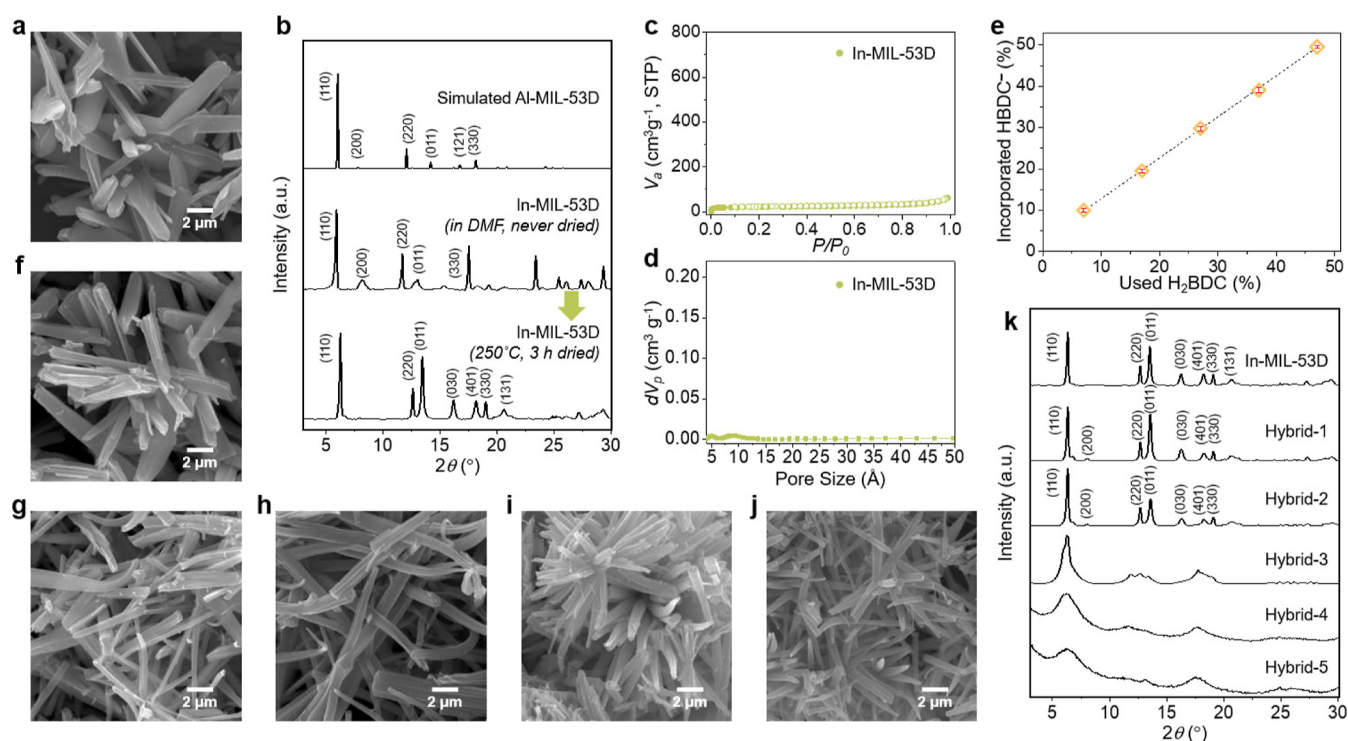


Figure 1. (a) SEM image of In-MIL-53D dried at 250 °C for 3 h. (b) PXRD patterns of simulated Al-MIL-53D, In-MIL-53D soaked in DMF (never dried), and In-MIL-53D dried at 250 °C for 3 h under vacuum. (c) N_2 sorption isotherm of In-MIL-53D dried at 250 °C for 3 h, displaying almost nonporous characteristics; the filled and open symbols represent the adsorption and desorption branches, respectively. (d) The pore-size distribution of In-MIL-53D calculated using the NLDFT method. (e) Ratios of the H_2BDC used in the solvothermal reactions of a series of defect-containing In-MIL-53D hybrids and ratios of the incorporated H_2BDC as determined by analyzing the 1H NMR spectra of the samples. Error bars represent data from four independent experiments. SEM images of (f) Hybrid-1, (g) Hybrid-2, (h) Hybrid-3, (i) Hybrid-4, and (j) Hybrid-5. (k) PXRD patterns of In-MIL-53D and a series of defect-containing In-MIL-53D hybrids.

MIL-53, a well-studied flexible MOF synthesized by reacting dicarboxylic acids with trivalent metal ions, features one-dimensional (1D) diamond-shaped channels that impart structural flexibility.^{32–35} An analog, MIL-53D (also known as DUT-5), can be synthesized by reacting 4,4'-biphenyldicarboxylic acid (H_2BPDC) and trivalent metal ions. Recently, we synthesized an indium-based variant of MIL-53D (In-MIL-53D).³² This MOF displayed minimal porosity upon drying, presenting challenges for its adsorption applications. This is a typical problem for flexible MOFs and an effective strategy must be developed to overcome this limitation. Herein, we demonstrate an approach for creating structural defects within the nearly nonporous In-MIL-53D framework by incorporating a shorter organic linker, which optimizes the gate-opening of the channels and the adsorption characteristics of the MOF. A series of hybrid MOFs incorporating defined ratios of H_2BPDC and 1,4-benzenedicarboxylic acid (H_2BDC as a shorter linker) were synthesized, and N_2 and CO_2 adsorption measurements were conducted for Brunauer–Emmett–Teller (BET) analysis. The judicious incorporation of H_2BDC into the In-MIL-53D framework created structural defects, activated gate-opening, and generated open metal sites. However, an excessive number of defects undermined the framework's integrity, compromising its stability and heightening the risk of collapse. Thus, it was crucial to attain an optimal defect level to maintain structural stability while enhancing functionality. These structural and chemical modifications of the MOFs result in a significant enhancement of CO_2 adsorption with up to an 11-fold increase in adsorption capacity at 1 P/P_0 compared to that of the pristine MOF. Enhanced adsorption performance was also confirmed

during the capture of chemical warfare agent (CWA) simulant. In-MIL-53D hybrids exposed to 2-chloroethyl ethyl sulfide (CEES) vapor in a jar-in-jar setup showed a steady increase in the adsorption with increasing numbers of structural defects, with up to 5-fold enhanced adsorption performance for CEES compared to that of pristine In-MIL-53D. By adjusting the defect levels, both the pore structure and the adsorption capacity can be fine-tuned to obtain a hybrid MOF with optimal performance for applications requiring enhanced porosity and structural flexibility.

RESULTS AND DISCUSSION

In-MIL-53D with a chemical composition of $[In(OH)(BPDC)]_n$ was synthesized via a solvothermal reaction of $In(NO_3)_3$ and H_2BPDC . In-MIL-53D is a three-dimensional (3D) orthorhombic MOF characterized by a rhombic lattice on the ab plane (Scheme 1b). The In-MIL-53D structure is composed of 1D metal-ion chains, specifically $[In(OH)]_n$ units aligned along the c -axis (Scheme 1b). Organic linkers extend from these chains in four directions, parallel to the ab plane and perpendicular to the 1D metal-ion chains. Ultimately, each linker connects to an adjacent 1D metal-ion chain, resulting in diamond-shaped 1D channels (Scheme 1b). This configuration is responsive to guest molecules and contributes to the flexibility of the MOF. Scanning electron microscopy (SEM) image and powder X-ray diffraction (PXRD) pattern of the product confirmed the formation of In-MIL-53D rods (Figure 1a,b). The unit cell parameters of the product (Table 1), calculated from the PXRD pattern by whole pattern fitting (Figure S1) using MDI Jade 9.0 software, revealed a 3D orthorhombic

Table 1. Unit Cell Parameters of Al-MIL-53D, In-MIL-53D, a Series of Defect-Containing In-MIL-53D Hybrids, and In-MIL-68B^c

MOF	cell parameters			δ^a (°)
	<i>a</i> (Å)	<i>b</i> (Å)	<i>c</i> (Å)	
Al-MIL-53D ³⁷	22.70	19.24	6.61	80.57
In-MIL-53D ^b	22.07	21.17	7.14	87.61
In-MIL-53D	25.46	16.70	7.11	66.53
Hybrid-0.5	26.14	16.50	7.22	64.52
Hybrid-1	26.08	16.46	7.17	64.53
Hybrid-1.5	26.12	16.50	7.23	64.56
Hybrid-2	26.02	16.49	7.21	64.73
Hybrid-2.5	25.11	16.84	7.06	67.68
In-MIL-68B	21.77	37.70	7.23	60.01
In-MIL-68B ³⁸	21.77	37.68	7.23	60.05

^aCalculated based on the cell parameters and confirmed using the structures obtained using Vesta software.³⁶ ^bIn-MIL-53D soaked in DMF, never dried. ^cUnit cell parameters were calculated by whole pattern fitting using MDI Jade 9.0 software. The angles (δ) between the organic linkers were calculated using the unit cell parameters. All samples were dried at 250 °C for 3 h under vacuum before analyzing, unless otherwise noted.

structure identical to that of the previously reported Al-MIL-53D. In particular, the product adopted an open form with *a* and *b* cell parameters at 22.07 and 21.17 Å, respectively, with a large angle of δ (87.61°, Scheme 1b), defined as the angle between the two linkers, before the evacuation process for removing the guest molecules located within the channels. However, the structure of In-MIL-53D changed to the closed form, after the evacuation process of the guest molecules with a dynamic vacuum for 3 h at 250 °C. The whole pattern fitting (Figure S2) revealed an increased *a* cell parameter at 25.46 Å, a decreased *b* cell parameter at 16.70 Å, and a small δ at 66.53°, which are representative of its closed form (Scheme 1b). Importantly, in its dried, closed form, In-MIL-53D displayed a nearly nonporous behavior in the N₂ sorption isotherm analysis (Figure 1c,d). The transition from an open to a closed structure during the guest evacuation process is a typical characteristic of flexible MOFs.^{25–35} Once dried, these MOFs adopt a closed form and sometimes cannot return to their open form, limiting their suitability for applications that require sustained flexibility (Scheme 1a). This behavior highlights the need for in-depth studies on flexible MOFs to better understand the processes involved in reopening so that their full potential can be realized in practical applications.

To activate the gate-opening behavior of In-MIL-53D for enlarged pore volume and enhanced adsorption performance, we employed a defect-engineering strategy by introducing H₂BDC, a shorter linker than the original H₂BPDC, during the synthesis of In-MIL-53D. The incorporation of this shorter linker created structural defects in the In-MIL-53D framework (Scheme 1b), facilitating gate-opening and improving the adsorption capacity, which was otherwise constrained in the pristine closed structure. A schematic representation of the defect-containing In-MIL-53D structure is shown in Scheme 1b. The structural conformation of MIL-53, whether in a closed or open form, is governed by two key factors: long-range dispersion interactions (π – π interactions between organic linkers) and vibrational entropy.^{33,39–41} Both experimental and theoretical studies suggest that π – π interactions between organic linkers stabilize the closed form of MIL-53, while the open form is

favored when entropy effects dominate.^{33,39–41} Therefore, weakening or reducing the π – π interactions between organic linkers within MIL-53 would destabilize the closed form and facilitate the gate-opening. The partial replacement of the longer BPDC^{2–} linkers with shorter HBDC[–] linkers will reduce these π – π interactions due to the decreased number of aromatic rings within the structure. The incorporation of shorter linkers into the framework also increased the number of open metal sites, enhancing the accessibility of the adsorption sites. Controlled defect incorporation within the MOFs was achieved through a series of solvothermal reactions by varying the ratio of the two linkers, H₂BPDC and H₂BDC, with In(NO₃)₃ (Scheme 1b). ¹H NMR spectroscopic analysis of the In-MIL-53D hybrids digested in acetic acid-*d*₄ and dimethyl sulfoxide (DMSO)-*d*₆ confirmed the incorporation ratios of the two linkers in the In-MIL-53D hybrids and showed that the short linker (H₂BDC) was incorporated into the framework with ca. 10, 20, 30, 39, and 50% (denoted as Hybrid-1–Hybrid-5, respectively). The ratios of H₂BDC used in the solvothermal reactions and the ratios of the incorporated short linkers within Hybrid-1–Hybrid-5 are depicted in Figure 1e (see also Table S1), indicating that the incorporated amount of the short linker could be precisely controlled. The reproducibility and controllability of the production of the In-MIL-53D hybrids were validated through four independent experiments. Each experiment yielded small error bars, demonstrating the precise control that was achieved for the incorporation of the two linkers within each hybrid, thereby allowing control over the number of defects and open metal sites. Additional analysis was performed to characterize defect formation. IR spectra of a series of samples confirmed the presence of defects within the frameworks. Specifically, the partial substitution of BPDC^{2–} with HBDC[–] resulted in the uncoordinated –CO₂H group and open metal sites. Consequently, in the In-MIL-53D hybrids, bare –CO₂H group should be incorporated into the framework. The presence of –CO₂H group⁴² in the In-MIL-53D hybrids was verified by IR spectra (Figure S3).

SEM images of the products of these reactions revealed rod-like structures similar to those of pure In-MIL-53D (Figure 1f–j). No signs of phase separation or production of byproducts, such as the distinct morphological characteristics of In-MIL-68B produced from In³⁺ and H₂BDC (Figure S4), were observed. PXRD analysis further confirmed these observations, as no peaks corresponding to In-MIL-68B were detected (Figure 1k). In particular, the strong peaks characteristic of In-MIL-68B at 4.69 and 9.38° for the (110) and (220) planes, respectively, were absent. For hybrids with up to 20% incorporation of the short linker (Hybrid-1 and Hybrid-2), the PXRD pattern remained nearly identical to that of the pristine form, indicating that the crystallinity of the framework was mostly preserved. In addition, the unit cell parameters of Hybrid-1 and Hybrid-2 (Table 1), calculated from their PXRD patterns by whole pattern fitting (Figures S5 and S6), were similar to those of pristine In-MIL-53D. However, as the incorporation of the short linker increased beyond this point (Hybrid-3–Hybrid-5), many defects and open metal sites were created, gradually affecting the overall crystallinity. This change was reflected in a decrease in the PXRD peak intensity and by peak broadening, suggesting slight destabilization of the framework. These results indicate that the incorporation of an excessive amount of the short linker can compromise the stability of In-MIL-53D, presumably because a higher proportion of the short linker disrupts the original framework, leading to structural instability and potential

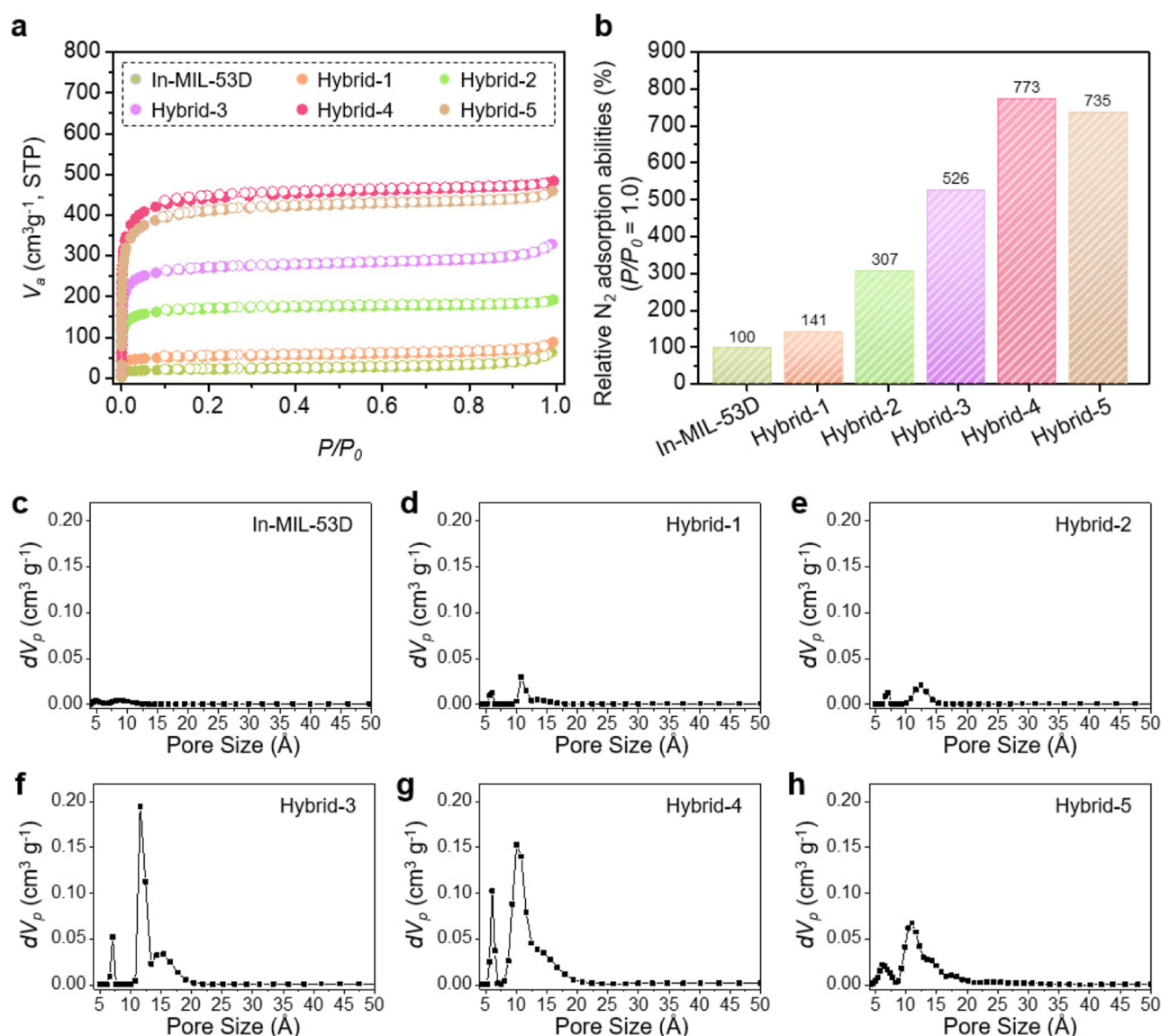


Figure 2. (a) N_2 sorption isotherms and (b) relative N_2 adsorption abilities (%) of In-MIL-53D and a series of defect-containing In-MIL-53D hybrids; the filled and open symbols represent the adsorption and desorption branches, respectively. In-MIL-53D (khaki), Hybrid-1 (orange), Hybrid-2 (light green), Hybrid-3 (purple), Hybrid-4 (red), and Hybrid-5 (brown). (c–h) Pore-size distributions of samples calculated using the NLDFT method.

collapse. This analysis indicates a pyramidal trend in performance, where an optimal level of structural defects can be identified that balances structural stability and enhanced functionality.

To quantify the impact of the incorporation of the short linker on structural recovery to its open form, N_2 sorption isotherms were measured after pretreatment at 250°C for 3 h (Figure 2a). By introducing defects, we aimed to facilitate the gate-opening behavior of the flexible MOF that initially persisted in their closed forms. As expected, the N_2 sorption isotherms increased as In-MIL-53D containing more of the short linker were used, reaching the highest values for Hybrid-4. Specifically, a BET surface area and total pore volume of $77.7 \text{ m}^2 \text{g}^{-1}$ and $0.097 \text{ cm}^3 \text{g}^{-1}$, respectively, were measured for pristine In-MIL-53D (Table S2). As the amount of short linker incorporation increased, the BET surface areas and total pore volumes of Hybrid-1 ($196.2 \text{ m}^2 \text{g}^{-1}$ and $0.134 \text{ cm}^3 \text{g}^{-1}$), Hybrid-2 ($649.1 \text{ m}^2 \text{g}^{-1}$ and $0.295 \text{ cm}^3 \text{g}^{-1}$), Hybrid-3 ($1033.7 \text{ m}^2 \text{g}^{-1}$ and $0.509 \text{ cm}^3 \text{g}^{-1}$), and Hybrid-4 ($1680.2 \text{ m}^2 \text{g}^{-1}$ and $0.747 \text{ cm}^3 \text{g}^{-1}$) showed a steady increase

(Table S2). However, Hybrid-5, with 50% incorporation of the short linker, displayed a slight decrease in BET surface area and total pore volume ($1534.8 \text{ m}^2 \text{g}^{-1}$ and $0.710 \text{ cm}^3 \text{g}^{-1}$) compared to the highest values at Hybrid-4. Thus, the incorporation of too many short linker units introduces severe structural instability in the MOFs and results in a decrease in porosity.

The relative N_2 adsorption capacities at 1.0 atm across the In-MIL-53D hybrid series exhibited a consistent upward trend, peaking at Hybrid-4 (Figure 2b). This enhanced performance, up to 773% higher than that of pristine In-MIL-53D, indicates that approximately 39% incorporation of the short linker is optimal for this series of MOFs. To further understand the observed N_2 sorption behavior, nonlocal density functional theory (NLDFT) was used to analyze the pore-size distributions of the In-MIL-53D hybrids. As expected, pristine In-MIL-53D exhibited minimal porosity (Figure 2c). However, with the incorporation of the short linker, the pore volume increased (Figure 2d–g). With the introduction of excessive defects, the overall pore volume began to decrease (Figure 2h). These

findings highlight the critical balance between defect incorporation and framework stability for optimizing the porosities of MOFs. By adjusting the defect levels, both the gate-opening and porosity can be fine-tuned to achieve a hybrid MOF with optimal performance for applications requiring enhanced porosity and structural flexibility.

The enhanced adsorption performance of the defect-engineered In-MIL-53D hybrids achieved by facilitating gate-opening was further validated by CO₂ adsorption measurements (Figure 3a). Overall, the defect-containing In-MIL-53D hybrids

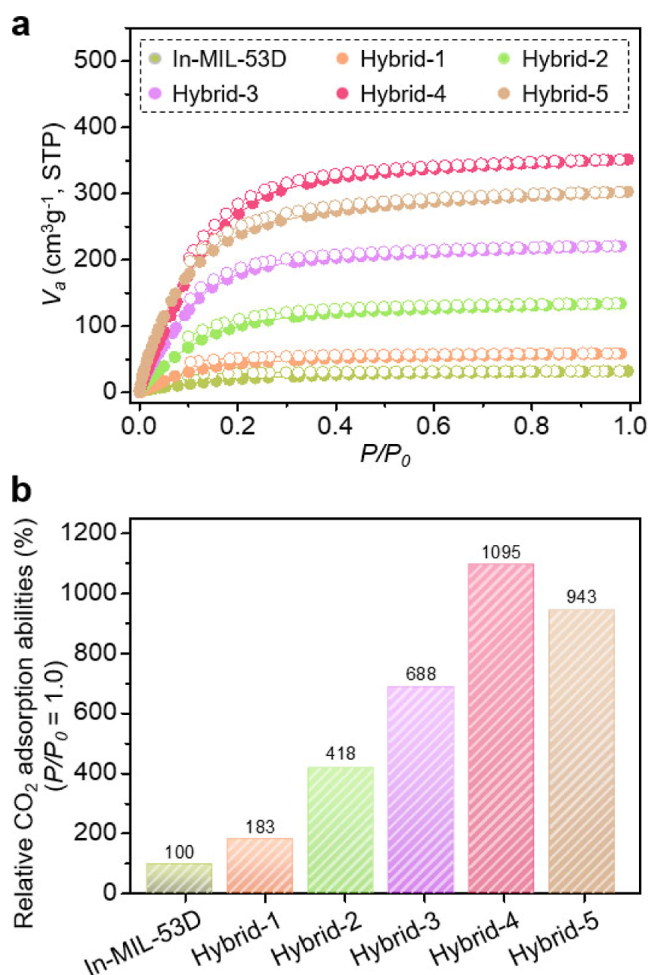


Figure 3. (a) CO₂ sorption isotherms and (b) relative CO₂ adsorption abilities (%) of In-MIL-53D and a series of defect-containing In-MIL-53D hybrids; the filled and open symbols represent the adsorption and desorption branches, respectively. In-MIL-53D (khaki), Hybrid-1 (orange), Hybrid-2 (light green), Hybrid-3 (purple), Hybrid-4 (red), and Hybrid-5 (brown).

exhibited a significant improvement in CO₂ adsorption capacity. Across the In-MIL-53D hybrid series, CO₂ adsorption displayed a consistent upward trend, peaking at Hybrid-4 before declining slightly at Hybrid-5 (Figure 3b), mirroring the behavior observed in the N₂ sorption isotherms. The maximum CO₂ uptake values for the In-MIL-53D hybrids are summarized in Table S3. Pristine In-MIL-53D exhibited a CO₂ uptake of 32.1 cm³ g⁻¹, while the modified hybrids—Hybrid-1 (58.8 cm³ g⁻¹), Hybrid-2 (134.3 cm³ g⁻¹), Hybrid-3 (220.9 cm³ g⁻¹), and Hybrid-4 (351.5 cm³ g⁻¹)—showed a consistent and progressive increase in CO₂ uptake (Table S3). A slight decrease was observed for Hybrid-5, with the CO₂ uptake measured at

302.6 cm³ g⁻¹. Notably, Hybrid-4 exhibited an almost 11-fold increase in CO₂ adsorption capacity at 1.0 atm compared to pristine In-MIL-53D (Figure 3b). The highly enhanced CO₂ adsorption capacity of the In-MIL-53D hybrids suggests that the incorporation of the short linker not only activates the gate-opening behavior of the nearly nonporous flexible In-MIL-53D framework but also optimizes its adsorption functionality by providing open metal sites. In general, the open metal sites in MOFs behave as Lewis acid sites, making them particularly advantageous for CO₂ adsorption.^{43–45} This approach significantly enhanced the CO₂ adsorption capacity, achieving levels comparable to those of highly porous MOFs (Table S4). Furthermore, the equilibration time for each data point during CO₂ adsorption was measured, as shown in Figure S7. For pristine In-MIL-53D, a sharp peak at very low CO₂ pressure was observed, indicating its nonporous nature and retention in the closed form. The most of flat region suggests that no more critical CO₂ adsorption because it had already reached saturation.^{29,46} In contrast, Hybrid-4 exhibited a prolonged equilibration time for CO₂ adsorption during approximately 0.0–0.3 P/P_0 , before reaching a flat region, indicating a gradual pore change and gate-opening during this CO₂ pressure range. To determine the CO₂ adsorption enthalpies in pristine In-MIL-53D and Hybrid-4, the isosteric heat of adsorption (Q_{st}) was calculated from adsorption isotherms measured at different temperatures using the Clausius–Clapeyron eq (Figure S8).^{47–49} The Q_{st} value for Hybrid-4 (27.1 kJ mol⁻¹) was higher than that of pristine In-MIL-53D (24.1 kJ mol⁻¹), indicating a stronger CO₂ adsorption affinity in Hybrid-4.

Given the success of the incorporation of the short linker in facilitating the gate-opening of In-MIL-53D, we sought to determine the maximum adsorption capacity that could be achieved for the In-MIL-53D hybrids and assess the potential for precise control over gate-opening. A new series of In-MIL-53D hybrids were synthesized with intermediate ratios between those previously established (Figure S9); thus, the incorporation of the short linker into the framework with proportions of approximately 5, 15, 25, 36, and 45% (denoting as Hybrid-0.5–Hybrid-4.5, respectively) was undertaken and confirmed by ¹H NMR spectroscopic analysis. Four independent experiments were conducted for each sample to establish the reproducibility, confirming the intended linker ratios. The ratios of H₂BDC used in the reactions and the ratios of its incorporation are depicted in Figure 4a, indicating that the incorporation ratio of the two linkers is quite controllable. This level of control supports the reliability of our synthesis approach, indicating the ability for the fine-tuning of the MOF structure. Such control over the MOF structure through deliberate defect incorporation may provide new opportunities for the optimization of flexible MOFs for gas storage and separation and for catalysis applications.

The SEM images of this new series show the formation of rod-like structures (Figure S9), consistent with the expected morphology. With up to 25% incorporation of the short linker (Hybrid-0.5–Hybrid-2.5), the PXRD pattern remained almost identical to that of the pristine form (Figure S9f), indicating that the crystallinity of the framework was mostly preserved. The unit cell parameters of these products (Table 1), calculated from their PXRD patterns by whole pattern fitting (Figures S10–S12), revealed that the values are similar to that of pristine In-MIL-53D. However, as the incorporation of the short linker increased beyond this point (Hybrid-3.5 and Hybrid-4.5), the PXRD peak intensity gradually decreased, and peak broadening was detected due to structural collapse, suggesting that a higher

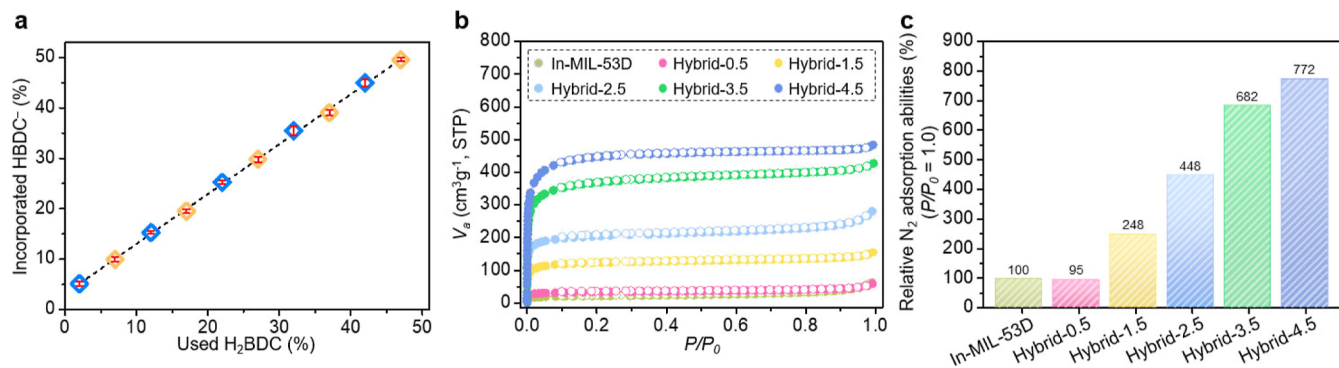


Figure 4. (a) Ratios of the H₂BDC used in the solvothermal reaction of a series of defect-containing In-MIL-53D hybrids and ratios of the incorporated HBDC⁻ as determined by analyzing the ¹H NMR spectra of the samples. The new series of hybrid samples are shown as blue dots. Error bars represent data from four independent experiments. (b) N₂ sorption isotherms and (c) relative N₂ adsorption abilities (%) of a series of defect-containing In-MIL-53D hybrids; the filled and open symbols represent the adsorption and desorption branches, respectively. In-MIL-53D (khaki), Hybrid-0.5 (pink), Hybrid-1.5 (yellow), Hybrid-2.5 (blue), Hybrid-3.5 (green), and Hybrid-4.5 (navy).

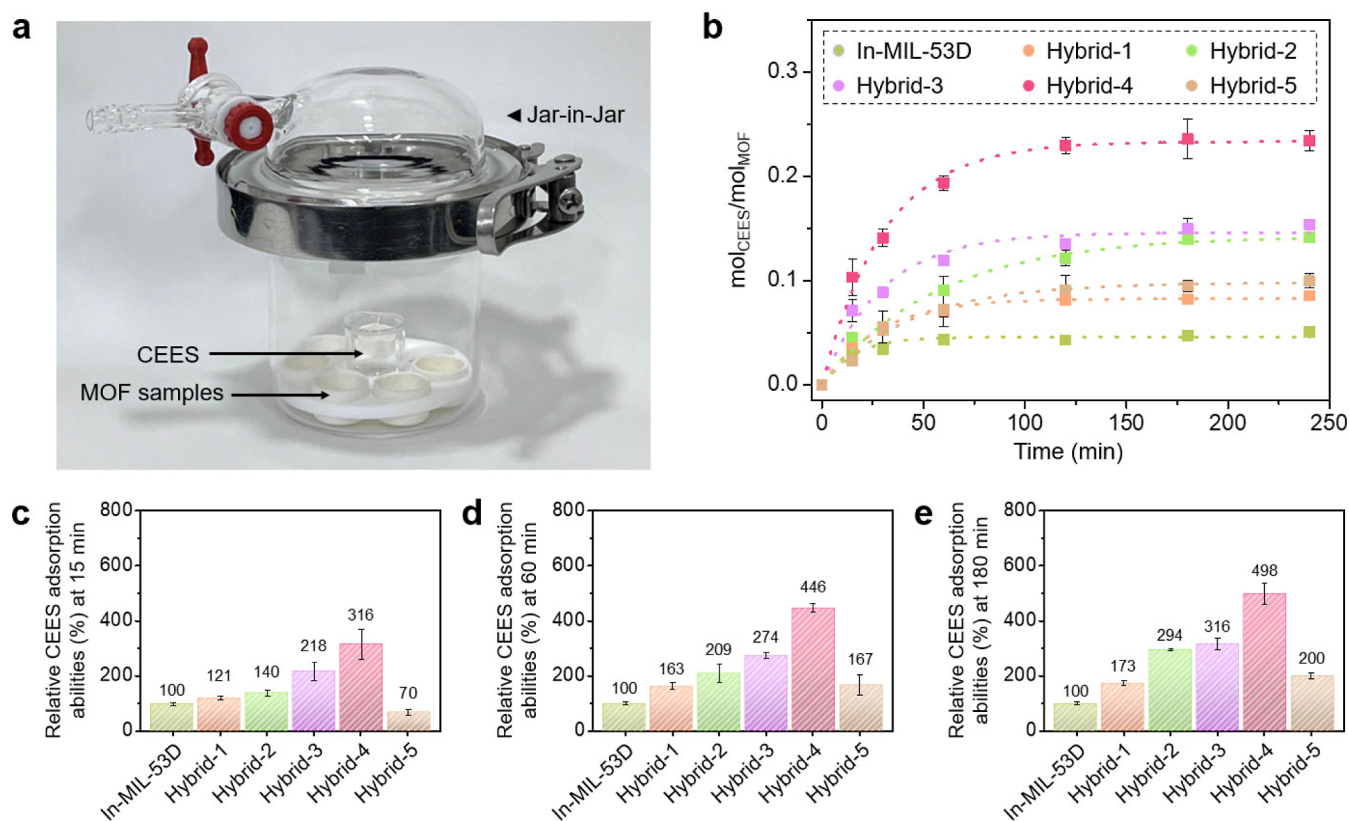


Figure 5. (a) Photograph of a custom-made jar-in-jar setup for the adsorption experiments with CEES. (b) CEES adsorption kinetics of In-MIL-53D and a series of defect-containing In-MIL-53D hybrids with the pseudo-first-order model. Relative CEES adsorption abilities (%) of In-MIL-53D and a series of defect-containing In-MIL-53D hybrids at (c) 15 min, (d) 60 min, and (e) 180 min. Error bars represent data from three independent experiments. In-MIL-53D (khaki), Hybrid-1 (orange), Hybrid-2 (light green), Hybrid-3 (purple), Hybrid-4 (red), and Hybrid-5 (brown).

content of the short linker begins to affect the structural integrity when it comprises one-fourth or more of the total linker ratio. This observation reinforces the importance of balanced defect incorporation for maintaining both structural stability and functional performance. N₂ sorption isotherms of samples (Figure 4b,c) revealed that Hybrid-0.5 (124.4 m² g⁻¹ and 0.091 cm³ g⁻¹), Hybrid-1.5 (467.4 m² g⁻¹ and 0.238 cm³ g⁻¹), Hybrid-2.5 (776.2 m² g⁻¹ and 0.433 cm³ g⁻¹), and Hybrid-3.5 (1373.9 m² g⁻¹ and 0.657 cm³ g⁻¹), and Hybrid-4.5 (1651.2 m² g⁻¹ and 0.746 cm³ g⁻¹) displayed a gradual increase in BET surface area and total pore volume (Table S2). However, the values for

Hybrid-4.5 are slightly lower than those of Hybrid-4. Two primary conclusions can be drawn from the experiments. First, maximizing the gate-opening in the nonporous In-MIL-53D is feasible by reacting In(NO₃)₃ with H₂BPDC and adding a short linker (H₂BDC) to achieve approximately 39% incorporation of the short linker. Second, precise control over the mixed linker content within the MOF framework allows the creation of a flexible MOF with an adjustable gate-opening, which is intrinsically linked to its adsorption capacity.

To evaluate the applicability of this approach to other flexible MOFs, Fe-MIL-88C with the chemical composition of [Fe₃O-

$(\text{NDC})_3(\text{H}_2\text{O})_2(\text{NO}_3)_n$ was synthesized via a solvothermal reaction of $\text{Fe}(\text{NO}_3)_3$ and 2,6-naphthalenedicarboxylic acid (H_2NDC).^{30,32} Additionally, two Fe-MIL-88C hybrids with controlled defect incorporation were prepared by varying the ratio of H_2NDC and H_2BDC in solvothermal reactions with $\text{Fe}(\text{NO}_3)_3$. All samples were characterized using ^1H NMR, SEM, and PXRD. SEM images verified the formation of hexagonal rods, characteristics of MIL-88C^{30,32} (Figure S13). ^1H NMR analysis of two Fe-MIL-88C hybrids verified the incorporation of the short linker (H_2BDC) into the framework at approximately 12 and 26%, respectively. Interestingly, PXRD patterns of the Fe-MIL-88C hybrids showed peak shifts compared to pristine Fe-MIL-88C, due to short linker incorporation, defect generation, and facilitated framework flexibility (Figure S13d). The unit cell parameters of the Fe-MIL-88C hybrids, determined from PXRD patterns via whole pattern fitting (Figures S14–S16) using MDI Jade 9.0 software, showed an increase in the *a* and *b* cell parameters and a decrease in the *c* cell parameter compared to those of pristine Fe-MIL-88C (Table S5). However, N_2 sorption isotherms indicated no significant enhancement in pore volume for the defected Fe-MIL-88C hybrids (Figure S13e).

Having demonstrated that MOFs with activated gate-opening, increased porosity, and enhanced site accessibility can be effectively engineered, we assessed their potential for removing toxic chemicals. The release of CWAs poses an immediate and severe risk to human health and safety.^{5–9} Rapid detection and removal of these toxic agents from the air are essential to prevent exposure because CWAs can cause debilitating effects, including severe skin rashes, respiratory damage, and even fatality. The gate-opening behavior of the In-MIL-53D hybrids shows promise for applications requiring responsive adsorption. Unlike pristine In-MIL-53D, which tends to remain closed once its gate is closed, the incorporation of a short linker into the In-MIL-53D framework enables a persistent gate-opening. This structural adaptation suggests that the In-MIL-53D hybrids can provide continuous and enhanced adsorption, making them highly suitable for real-time CWA removal from the environment. Due to the hazards associated with actual CWAs, CEES was selected as a safe laboratory test agent for the CWA adsorption experiments. CEES is a well-known simulant of sulfur mustard (bis(2-chloroethyl)sulfide), which is a typical CWA that can cause severe tissue damage upon exposure.^{6–9} The relative CEES adsorption capabilities of pristine In-MIL-53D and its hybrid variants were evaluated using the jar-in-jar setup at 23 °C (Figure 5a). In this setup, a small beaker containing CEES was placed at the center of a larger glass chamber, with six crucible pans containing the MOF samples positioned around, thereby exposing the samples to CEES vapor for defined time intervals.

The adsorption of CEES within the MOF samples was confirmed using both energy-dispersive X-ray spectroscopy (EDX) and ^1H NMR spectroscopy. The EDX spectra of the samples revealed the presence of sulfur and chlorine atoms in the adsorbed CEES, whereas neither was detected in the samples before the CEES adsorption experiment (Figures S17 and S18). To quantify the amount of CEES adsorbed by each sample, the samples exposed to CEES vapor for various time intervals were dissolved in acetic acid- d_4 and DMSO- d_6 for ^1H NMR spectroscopic analysis. The number of CEES molecules adsorbed per In-MIL-53D unit was calculated by comparing the integrated ^1H NMR peaks of CEES with those of the organic linkers from In-MIL-53D (Figure 5). The quantities of CEES

adsorbed at different exposure times are plotted in Figure 5b. After reaching saturation following 3 h of exposure to CEES vapor, the adsorption capacity of pristine In-MIL-53D was determined to be 0.05 mol of CEES per mole of In-MIL-53D (0.05 mol_{CEES}/mol_{MOF}) (Table S6). All In-MIL-53D hybrids displayed enhanced CEES adsorption capacities compared to that of pristine In-MIL-53D, both during the early stage of CEES exposure and after saturation (Figure 5c–e). The increase in CEES adsorption capacity occurred in a stepwise, progressive manner. However, a steep increase was observed with the hybrid containing approximately 39% incorporation of the short linker, which achieved approximately 5-fold higher adsorption capacity relative to pristine In-MIL-53D, reaching 0.24 mol_{CEES}/mol_{MOF} after 3 h exposure (Figure 5b and Table S6). This dramatic enhancement highlights the effective combination of the activated gate-opening and additional open metal sites, resulting in the maximized adsorption capacity of Hybrid-4. The adsorption kinetics of pristine In-MIL-53D and its hybrids are illustrated in Figure 5 and summarized in Table S7. The adsorption of CEES onto In-MIL-53D and a series of defect-containing In-MIL-53D hybrids followed a pseudo-first-order kinetic model.

CONCLUSION

Flexible MOFs with deliberately introduced defects were synthesized using a ligand-mixing strategy. The samples exhibited enhanced porosity and increased adsorption capacity. By introducing a shorter organic linker into the nearly nonporous In-MIL-53D framework, structural defects were introduced that optimized the channel gate-opening and improved the adsorption performance. A series of ten In-MIL-53D hybrids were prepared with precisely controlled ratios of the original linker (H_2BPDC) and the shorter linker (H_2BDC ; ranging from 5% to 50% in 5% intervals). The structural defects activated the gate-opening, significantly enhancing the gas adsorption capacities of the MOFs for N_2 and CO_2 . Additionally, the hybrids demonstrated excellent adsorption performance for the CWA simulant, CEES. The incorporation of a shorter linker created open metal sites, further contributing to improved adsorption. However, the incorporation of too many defects disrupted the framework's integrity and compromised its stability, leading to potential collapse. Thus, achieving an optimal level of defect incorporation is crucial for balancing structural stability and enhanced functionality. By fine-tuning the defect levels, both the pore structure and the adsorption capacity were optimized, resulting in a hybrid MOF with superior performance. Among the synthesized hybrids, the sample with approximately 39% short linker incorporation exhibited the best results, showing an 11-fold increase in CO_2 adsorption capacity at 1 *P/P*₀ and up to a 5-fold increase in CEES adsorption capacity compared to pristine In-MIL-53D. These enhancements arose from the synergistic effects of the activated gate-opening and additional open metal sites. The softened, flexible framework opens up more readily in response to environmental changes, highlighting the potential of these tailored hybrid MOFs for advanced applications. This study emphasizes the critical balance between defect incorporation and framework stability, demonstrating the versatility of engineered MOFs for diverse practical applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c01399>.

Materials, methods, experimental details, whole pattern fittings, IR spectra, SEM images, PXRD patterns, kinetics and thermodynamics data for CO₂ adsorption, EDX spectra, ratios of two linkers in In-MIL-53D hybrids, BET surface areas, total pore volumes, CO₂ uptake capacities, CEES adsorption capacities, kinetic parameters for CEES adsorption, analysis of Fe-MIL-88C hybrids, Figures S1–S18, Table S1–S7 (PDF)

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Notes

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